

An improved TODIM-TOPSIS method for quality evaluation of college students employment and entrepreneurship education with probabilistic hesitant fuzzy sets

Lijuan Zhao* and Shuo Du

Shijiazhuang University of Applied Technology, Shijiazhuang, Hebei, China

Abstract. In recent years, employers have continuously raised their requirements for college students, not only requiring a solid professional foundation, but also emphasizing personal professional literacy. As the first base for cultivating college students, major universities should not only guide them in their correct employment and entrepreneurship, but also help them find employment and entrepreneurship faster and better. However, in the context of the new era, universities still face some problems in the process of carrying out employment and entrepreneurship education, which hinder the progress of employment and entrepreneurship education. The probabilistic hesitant fuzzy sets (PHFSs), as an extension of hesitant fuzzy sets (HFSs), can more effectively and accurately describe uncertain or inconsistent information during the quality evaluation of college student employment and entrepreneurship education. TODIM and TOPSIS methods are two commonly used multi-attribute decision-making (MADM) methods, each of which has its advantages and disadvantages. The quality evaluation of college student employment and entrepreneurship education is regarded as the defined multiple attribute group decision making (MAGDM). This paper proposes a novel method based on TODIM and TOPSIS to cope with multi-attribute group decision making (MAGDM) problems under PHFSs environment. After introducing the related theory of PHFSs and the traditional TODIM and TOPSIS methods, the novel method based on a combination of TODIM and TOPSIS methods is designed. And then, an illustrative example for quality evaluation of college student employment and entrepreneurship education proved the feasibility and validity of the proposed method. Finally, the result has been compared with some existing methods under the same example and the proposed method's superiority has been proved.

Keywords: Multi-attribute group decision making (MAGDM), probabilistic hesitant fuzzy sets (PHFSs), TODIM method, TOPSIS method, employment and entrepreneurship education

1. Introduction

Multi-attribute decision making (MADM) is a modern decision science, and one of the important parts of that is that a decision maker (DM) is based on multiple attributes to choose an optimal solution [1–6]. Considering the practical decision-making circumstance, the decision-making problems always

need to judge from multi-view [7–9]. In daily life, when faced with relatively complex problems, it is difficult for people to directly give accurate values to give rational judgments in line with the objective reality [10–15]. In most cases, they can only express preferences in uncertain language based on their subjective understanding of evaluation schemes [16–20]. Therefore, it is necessary to study various linguistic information models. In order to make decisions with the fuzzy description, Zadeh [21] introduced fuzzy sets, and now it's used for the research and

*Corresponding author. Lijuan Zhao. E-mail: zhaolijuan202306@163.com.

decision making and control [22–25]. Then the extension of fuzzy set for instance, intuitionistic fuzzy set was proposed by Atanassov [26] and hesitant fuzzy sets (HFSs) was then proposed by Torra, Narukawa and Ieee [27] to extend the fuzzy set. In the hesitant fuzzy environment, the membership degree could be indicated by some possible number. On this basis, the probabilistic hesitant fuzzy sets [28] (PHFSs) are introduced. The PHFS could contain more information than other fuzzy circumstance. Then the distance evaluation and the score function are the proposed to put it better use of the PHFS. In addition to the expansion of the PHFS' own calculation rules, the PHFS is combined with a variety of fuzzy decision methods and decision theories to better simulate the actual decision environment. Guo, Yin, Zhang, Lin and Li [29] utilized the PHFE to calculate the hamming distance and used the extended TODIM method to evaluate the CCUS storage site. Chen, Shen, Ren, Yu and Yuan [30] integrated GRA and TOPSIS in PHFS to evaluate the airspace operation effectiveness. Li [31] constructed PHF-VIKOR model to evaluate the management quality of teacher education.

With the increasing enrollment expansion of major universities, the employment situation for college students is becoming increasingly severe. Carrying out employment and entrepreneurship education has become an effective measure to alleviate the current employment pressure on college students [32–34]. Carrying out employment and entrepreneurship education in the context of the new era is of great significance. It not only helps guide college students to establish employment concepts and alleviate employment pressure, but also helps to promote higher education reform and improve their competitiveness. Due to the fact that the number of graduates each year is far greater than the demand for social positions, and the current talent market is also relatively saturated, many college students have difficulty finding employment [35–37]. Conducting employment and entrepreneurship education in schools can expand students' career cognition, enable them to understand job positions closely related to their majors, formulate personal career plans, and strive towards this goal. Moreover, some college students often have incorrect employment concepts such as high expectations, low skills, and blind employment in the early stages of employment. Using employment and entrepreneurship education can help college students establish correct employment concepts, avoid and eliminate some incorrect employment concepts, and thus help them establish healthy and correct

employment concepts, find suitable job positions, and achieve faster and better employment [38, 39]. Although the number of university enrollment in China has significantly increased in recent years, some universities still face the problem of difficult enrollment. The reasons for this are mainly twofold: firstly, the poor teaching quality of schools themselves, and secondly, the low employment rate of school graduates. Employment and entrepreneurship education is driving the reform of higher education, transforming the way universities cultivate talents from focusing on professional skills to cultivating talents that are in line with the development of the employment situation [40, 41]. In terms of improving employment rates, employment and entrepreneurship education can not only improve the overall employment rate of college students, but also guide students to find their own career development plans in practical education and teaching. College students receive high-quality teaching and education in school, and find good jobs after graduation, which is the most effective way to improve the competitiveness of the school itself [42, 43]. In order to help college students find employment better and faster, major universities have increased their emphasis on the quality of employment and entrepreneurship education, invested a large amount of educational resources, and also achieved certain results [44, 45]. However, we can also see that currently, college students' employment and entrepreneurship education still faces some problems, specifically manifested in the following three aspects. (1) In the new era of outdated teaching content and lack of innovative educational resources, using the Internet for employment and entrepreneurship teaching can provide vivid and systematic employment and entrepreneurship teaching for college students. For example, by analyzing employment and entrepreneurship cases on the Internet, it can help students sort out their knowledge of employment and entrepreneurship, enhance technical practice and ideological depth. In recent years, the development of employment and entrepreneurship education has made certain achievements, but some colleges and universities still have the problem of outdated teaching content, and there are few relevant courses using the "Internet plus" technology. Moreover, some of the teaching content still follows classic cases from the past, which cannot keep up with the current employment trend, resulting in a disconnect between the teaching content and the current employment and entrepreneurship situation. Moreover, whether it is university employment

and entrepreneurship course resources or related educational resources on the Internet, their form and content are mostly identical, lacking innovative employment and entrepreneurship resources [46, 47]. (2) The education faculty is insufficient, and the lack of professional guidance for employment and entrepreneurship education generally involves knowledge of multi-level analysis. Teaching teachers are crucial for theoretical understanding and practical experience, especially for analyzing the current employment and entrepreneurship situation rather than imparting theoretical knowledge. In Western countries, employment and entrepreneurship teachers in universities are mostly individuals with rich practical experience or unique insights in this field [48, 49]. They can combine theoretical knowledge with practical experience, thus providing students with vivid and personalized classrooms. However, in contrast to the current situation of employment and entrepreneurship education in our country, most teachers are not professional employment and entrepreneurship teachers, and some schools only allow ideological and political teachers to concurrently serve as this subject. Because they cannot fully understand the knowledge and theory of employment and entrepreneurship, and their practical experience is relatively scarce, it is easy to have dull teaching classrooms that cannot stimulate students' desire to learn about employment and entrepreneurship. Unable to effectively leverage the role of employment and entrepreneurship education [50]. (3) Emphasizing the cultivation of theoretical knowledge and lacking specific practical experience, employment and entrepreneurship education not only needs to cultivate students' solid theoretical knowledge, but also requires them to understand the principles through specific practice, find and improve their shortcomings in specific practice, and thus better employment and entrepreneurship. At present, most universities still have the problem of emphasizing theoretical knowledge cultivation and lacking specific practice in employment and entrepreneurship education. Internship training is the most commonly used employment and entrepreneurship practice method in most universities, but there are still some problems in the process of internship training. For example, some universities do not have clear teaching objectives for internship training, resulting in students not learning knowledge during the internship process [51, 52]. In addition, some universities only focus on students' internship certificates and neglect the supervision of the internship process, leading to the phenomenon

of some students issuing fake internship certificates. So when universities carry out employment and entrepreneurship education, they should change the traditional teaching mode, effectively combine theoretical knowledge cultivation with practical activities, increase the supervision mechanism of employment and entrepreneurship education, and promote the all-round development of college students. Internship for college students is an effective teaching method to help them accumulate employment and entrepreneurship experience and improve their hands-on practical skills [53–55]. During the internship process, college students can identify the gap between their own abilities and the talent standards of the enterprise, and then make good employment and entrepreneurship plans. Moreover, in order to effectively carry out employment and entrepreneurship education, major universities will integrate resources from schools, enterprises, and governments based on their own situation, and actively provide internship and practical enterprises for college students. However, some schools' teaching efforts in school enterprise cooperation are not strong enough, which is reflected in the uneven quality of internship enterprises, resulting in students not learning knowledge during the internship process and losing the significance of internship practice. On the other hand, it reflects the need for improvement in the teaching methods for college students' internships. Some universities have developed an educational approach that only focuses on the results of internships and neglects the process of internships, resulting in the formalization of internship practice [56, 57]. So major universities should select high-quality enterprises for college students to practice internships, strengthen the regulatory system during the internship process, and thereby improve the quality of college students' internships. In recent years, major universities have increased their teaching efforts in the quality of employment and entrepreneurship education, achieving some results in innovative teaching design and practical forms, and significantly improving teaching effectiveness [58, 59]. However, due to the drawbacks of traditional education models and the constantly changing employment situation, employment and entrepreneurship education still needs to be continuously improved and innovated. Therefore, in the context of the new era, major universities should continuously enrich and innovate their teaching content, improve the quality of practical internships for college students, and effectively alleviate the problem of difficult employment for college students [60, 61].

The probabilistic hesitant fuzzy sets (PHFSs), as an extension of hesitant fuzzy sets (HFSs), can more effectively and accurately describe uncertain or inconsistent information during the quality evaluation of college student employment and entrepreneurship education. TODIM and TOPSIS methods are two commonly used MADM methods, each of which has its advantages and disadvantages. The quality evaluation of college student employment and entrepreneurship education is regarded as the defined MAGDM. This paper proposes a novel method based on TODIM and TOPSIS to cope with the MAGDM problems under PHFSs environment. After introducing the related theory of PHFSs and the traditional TODIM and TOPSIS methods, the novel method based on a combination of TODIM and TOPSIS methods is designed. And then, an illustrative example for quality evaluation of college student employment and entrepreneurship education proved the feasibility and validity of the proposed method. Finally, the result has been compared with some existing methods under the same example and the proposed method's superiority has been proved.

The following is the full text of this article. Section 2 gives the Preparatory knowledge of PHFSs, Section 3 gives the TODIM-TOPSIS method for MAGDM under PHFSs. In Section 4, a case study for quality evaluation of college student employment and entrepreneurship education is used to experiment with the proposed method and some comparative analysis are given to prove this method's effectiveness and accuracy. At last, the Section 5 draws a conclusion of this article.

2. Preliminary knowledge

The hesitant fuzzy sets (HFs) are proposed to list all possible values as membership values. the HFs are shown as follows:

2.1. The HFSs and PHFSs

Definition 1 [62]. Assume a fixed set K , then the HFS R on K is defined as:

$$R_k = \{r_k(m_k^a | m_k^a)\} \quad (1)$$

where $r_k^a \in R$, $0 \leq m_k^a \leq 1$, $a = 1, 2, \dots, \#m$, and $\#m$ represents the value of elements.

However, the HFs defaults that all membership degrees in the set have the same importance in the

decision-making process, which leads to the expert's inability to reflect his own preferences in the evaluation. Thus, to solve this problem probabilistic hesitant fuzzy sets (PHFSs) have been developed which considers to add the corresponding possibility for membership.

Definition 2 [28]. Set a fixed aggregation K , then the PHFS R on K is defined as:

$$R_k = \{r_k(m_k^a | q_k^a) | m_k^a, q_k^a\} \quad (2)$$

where $r_k^a \in R$, $0 \leq m_k^a \leq 1$, $a = 1, 2, \dots, \#m$, and $\#m$ represents the elements value, m_k^a denotes the possible membership degree, q_k^a denotes the probability of the probabilistic hesitant fuzzy element (PHFE), and $\sum_{a=1}^{\#m} q_k^a = 1$. Since the PHFSs is faced with the product problem, the different probabilities of corresponding membership degrees between different sets will affect the calculation results, so a new standardization method is adopted [63]. The corresponding probabilities of all membership degrees in the matrix are adjusted uniformly before calculation.

Definition 3 [63]. Let $r(m_k | q_k) = \{m^k(q^k)\}$, $r_1(m_a | q_a) = \{m_1^a(q_1^a)\}$ and $r_2(m_b | q_b) = \{m_2^b(q_2^b)\}$ be three PHFEs, $k = 1, 2, \dots, \#m$, $a = 1, 2, \dots, \#m_1$, $b = 1, 2, \dots, \#m_2$.

Step 1. Determine the first part of the PHFE. If $q_1^1 < q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_1^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1)$, otherwise, $m_1^1(q_1^1) = m_1^1(q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1)$.

Step 2. Determine the second part of the PHFE. If $q_1^1 < q_2^1$ and $q_2^1 - q_1^1 \leq q_1^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1 - q_1^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1 - q_1^1)$. If $q_1^1 < q_2^1$ and $q_2^1 - q_1^1 > q_1^1$, then $m_1^1(q_1^1) = m_1^1(q_1^1)$ and $m_2^1(q_2^1) = m_2^1(q_1^1)$. If $q_1^1 \geq q_2^1$ and $q_1^1 - q_2^1 \leq q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_1^1 - q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_1^1 - q_2^1)$. If $q_1^1 \geq q_2^1$ and $q_1^1 - q_2^1 > q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1)$.

Step 3. Determine the third part of the PHFE. If $q_1^1 \geq q_2^1$, $q_1^1 - q_2^1 \leq q_2^1$ and $q_2^1 \leq q_2^1 - q_1^1 + q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1)$. If $q_1^1 \geq q_2^1$, $q_1^1 - q_2^1 \leq q_2^1$ and $q_2^1 > q_2^1 - q_1^1 + q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1 + q_2^1 - q_1^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1 + q_2^1 - q_1^1)$. If $q_1^1 \geq q_2^1$, $q_1^1 - q_2^1 > q_2^1$ and $q_2^1 \geq q_2^1 + q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1)$. If $q_1^1 \geq q_2^1$, $q_1^1 - q_2^1 > q_2^1$ and $q_2^1 < q_2^1 + q_2^1$, then $m_1^1(q_1^1) = m_1^1(q_2^1 - q_2^1)$ and $m_2^1(q_2^1) = m_2^1(q_2^1 - q_2^1)$. If $q_1^1 < q_2^1$,

$q_2^1 - q_1^1 \leq q_1^2$ and $q_1^2 + q_1^1 \leq q_2^2 + q_2^1$, then $m_1^3(q_1^3) = m_1^2(q_1^2 - q_2^1 + q_1^1)$ and $m_2^3(q_2^3) = m_2^2(q_1^2 - q_2^1 + q_1^1)$. If $q_1^1 < q_2^1$, $q_2^1 - q_1^1 > q_1^2$ and $q_2^1 + q_1^2 \leq q_1^3 + q_1^1$, then $m_1^3(q_1^3) = m_1^1(q_1^2 - q_1^1 - q_1^2)$ and $m_2^3(q_2^3) = m_1^2(q_2^1 - q_1^1 - q_1^2)$. If $q_1^1 < q_2^1$, $q_2^1 - q_1^1 > q_1^2$ and $q_2^1 + q_1^2 > q_1^3 + q_1^1$, then $m_1^3(q_1^3) = m_1^3(q_1^3)$ and $m_2^3(q_2^3) = m_2^1(q_1^3)$. where $q_1^1 + q_1^2 + \dots \cdot q_1^{\#m_1} = 1$ and $q_2^1 + q_2^2 + \dots \cdot q_2^{\#m_2} = 1$, $\#m_1 = \#m_2$.

Definition 4 [28]. Calculate the score function by Equation (3)

$$s(m(q)) = \sum_{k=1}^{\#m} m^k q^k \quad (3)$$

where m^k denote the k -th largest elements of PHFE, and q^k denotes the probability of m^k .

Definition 5 [64]. The comparison rules between $r_1(q_1) = \{m_1^a(q_1^a)\}$ and $r_2(q_2) = \{m_2^b(q_2^b)\}$ are shown as follows:

- (1) $\bar{r}_1(q_1) > r_2(\bar{q}_2)$, if $s(\bar{r}_1(q_1)) > s(r_2(q_2))$
- (2) $r_1(q_1) > \bar{r}_2(q_2)$, if $s(r_1(q_1)) > s(r_2(q_2))$ and $d(r_1(q_1)) < d(r_2(q_2))$
- (3) $r_1(q_1) = r_2(q_2)$, if $s(r_1(q_1)) = s(r_2(q_2))$ and $d(r_1(q_1)) = d(r_2(q_2))$
- (4) $r_1(q_1) < r_2(q_2)$, if $s(r_1(q_1)) = s(r_2(q_2))$ and $d(r_1(q_1)) > d(r_2(q_2))$

Definition 6 [64]. $r_1(q_1) = \{m_1^a(q_1^a)\}$ and $r_2(q_2) = \{r_2^a(q_2^a)\}$ are two standardized PHFEs, where $\#m_1 = \#m_2 = \#m$ and $q_1^a = q_2^b = q^k$. The Lance distance is calculated as Equation (4)

$$d(r_1(q_1), r_2(q_2)) = \frac{1}{\#m} \sum_{a=b=1}^{\#m} \frac{|m_1^a q_1^a - m_2^b q_2^b|}{m_1^a q_1^a + m_2^b q_2^b} \quad (4)$$

Definition 7 [63]. $r_1(q_1) = \{m_1^a(q_1^a)\}$ and $r_2(q_2) = \{m_2^b(q_2^b)\}$ are computed as follows.

$$(1) r_1(q_1) \oplus r_2(q_2) = \cup_{a=1, \dots, \#m_1, b=1, \dots, \#m_2} \{m_1^a + m_2^b - m_1^a m_2^b (q^j)\}; \quad (5)$$

$$(2) r_1(q_1) \otimes r_2(q_2) = \cup_{a=1, \dots, \#m_1, b=1, \dots, \#m_2} \{m_1^a m_2^b (q^k)\} \quad (6)$$

Definition 8 [65]. The PHF weighted averaging (PHFWA) operator is calculated by Equation (7)

$$\begin{aligned} PHFWA(r_1(q_1), r_2(q_2), \dots, r_l(q_l)) &= \bigoplus_{c=1}^l (r_c w_c) \\ &= \bigcup_{m_1 \in r_1, m_2 \in r_2, \dots, m_l \in r_l} \left\{ 1 - \prod_{c=1}^l (1 - m_c)^{w_c} (q^c) \right\} \end{aligned} \quad (7)$$

where $w_c = (w_1, w_2, \dots, w_l)$ is the weight and $\sum_{c=1}^l w_c = 1$, $w_c \in [0, 1]$.

Definition 9 [65]. The equation of PHF weighted geometric (PHFWG) operator is defined in Equation (8)

$$\begin{aligned} PHFWG(r_1(q_1), r_2(q_2), \dots, r_l(q_l)) &= \bigoplus_{c=1}^l (r_c)^{w_c} \\ &= \bigcup_{m_1 \in r_1, m_2 \in r_2, \dots, m_l \in r_l} \left\{ \prod_{c=1}^l (m_c)^{e_c} (q^c) \right\} \end{aligned} \quad (8)$$

2.2. Classical TODIM method

The core idea of TODIM method is derived from the prospect theory which compares the alternatives by the predominance degree of alternative over others. The combination of classical TODIM method and PHFSs [66] are shown as follows.

Suppose that there are m solutions $AL = (AL_1, AL_2, \dots, AL_m)$ to be selected, and n attributed $AT = (AT_1, AT_2, \dots, AT_n)$. The decision matrix is presented as $M = [r_{xy}(q_{xy})]_{m \times n}$, and the weighting vector is denoted as $w_y = (w_1, w_2, \dots, w_n)$. The main steps of classical TODIM method are listed as follows:

Step 1: Standardized the decision information.

Transform the cost attribute into benefit attribute by the following equations:

When AT_y is the positive attribute, the normalized PHFE is determined by Equation (9)

$$r_{xy}^* = \{m_{xy}(q_{xy})\} \quad (9)$$

When AT_y is the negative attribute, the normalized PHFE is determined by Equation (10)

$$r_{xy}^* = \{(1 - m_{xy})(q_{xy})\} \quad (10)$$

Then normalize the decision matrices using the steps in Definition 3.

Step 2: calculate the relative weight by Equation (11)

$$w_y^* = \frac{w_y}{\ddot{w}} \quad (11)$$

where w denotes the maximum value of the attributes' weight.

Step 3: Calculate the dominance degree of the alternative S_k over the other alternative

The dominance degree of each alternative is figured out as Equation (12)

$$\tau(S_k) = \tau(S_x, S_k) = \sum_{x=1}^m \sum_{y=1}^n \vartheta_{xky}(S_x, S_k) \quad (12)$$

the preference index number is computed by Equation (13)

$$\vartheta_{xky}(S_x, S_k) = \begin{cases} \sqrt{\frac{w_y^* \times d(r_{xy}, r_{ky})}{\sum_{y=1}^n w_y^*}} & \text{if } s(r_{xy}) \geq s(r_{ky}) \\ -\theta \sqrt{\frac{\sum_{y=1}^n w_y^* \times d(r_{xy}, r_{ky})}{w_y^*}} & \text{if } s(r_{xy}) < s(r_{ky}) \end{cases} \quad (13)$$

Finally, calculate the overall dominance degree of the alternative by Equation (14).

$$\varphi(S_x) = \frac{\tau(S_x) - \min(\tau(S_x))}{\max(\tau(S_x)) - \min(\tau(S_x))} \quad (14)$$

Step 4: the final ranking results are ranked by the overall dominance degree of alternatives in a descending order.

2.3. Classical TOPSIS method

The classical TOPSIS method is a MADM method based on geometric thinking by evaluating distance between possible solution and ideal point. Let $AL = (AL_1, AL_2, \dots, AL_m)$ be the alternatives, and $AT = (AT_1, AT_2, \dots, AT_n)$ be the attributes. The decision matrix is denoted as $M = [r_{xy}(q_{xy})]_{m \times n}$, and the weighting vector is denoted as $w_y = (w_1, w_2, \dots, w_n)$. The specific steps in PHFSs [67] are shown as follows:

Step 1: Standardize the decision matrix $M^* = [r_{xy}^*(q_{xy}^*)]_{m \times n}$ using Equations (15), (16)

Transform the cost attribute into benefit attribute by the following equations:

When AT_y is the positive attribute, the normalized PHFE is determined by Equation (15)

$$r_{xy}^* = \{m_{xy}(q_{xy})\} \quad (15)$$

When AT_y is the negative attribute, the normalized PHFE is determined by Equation (16)

$$r_{xy}^* = \{(1 - m_{xy})(q_{xy})\} \quad (16)$$

Then normalize the decision matrices using the steps in Definition 3.

Step 2: Determine the positive and negative ideal solution $(r^+_y(q^+_y))$ and $(r^-_y(q^-_y))$ indexes by the following equation:

$$\begin{cases} r^+_y(q^+_y) = \max_y s(r_{xy}^*) \\ r^-_y(q^-_y) = \min_y s(r_{xy}^*) \end{cases} \quad (17)$$

Step 3. Calculate the positive and negative distance

$$d_y^+(r_{xy}^*, r_{xy}^+) = \sum_{y=1}^n w_c \frac{1}{\#m} \sum_{a=b=1}^{\#m} \frac{|m_{xy}^a q_{xy}^a - m_{xy}^b q_{xy}^b|}{m_{xy}^a q_{xy}^a + m_{xy}^b q_{xy}^b} \quad (18)$$

$$d_y^-(r_{xy}^*, r_{xy}^-) = \sum_{y=1}^n w_c \frac{1}{\#m} \sum_{a=b=1}^{\#m} \frac{|m_{xy}^a q_{xy}^a - m_{xy}^b q_{xy}^b|}{m_{xy}^a q_{xy}^a + m_{xy}^b q_{xy}^b} \quad (19)$$

Step 4. Compute out the relative closeness to the ideal solution

The relative closeness of alternative Q_r is calculated by Equation (20)

$$H_x = \frac{d_y^-(r_{xy}^*, r_{xy}^-)}{d_y^+(r_{xy}^*, r_{xy}^+) + d_y^-(r_{xy}^*, r_{xy}^-)} \quad (20)$$

Step 5. Get the final ranking

The final rank is determined by the value of relative closeness, and the sort rule is the bigger the relative closeness is, the more appropriate the scheme is.

3. The TODIM-TOPSIS method for MAGDM under PHFSs

The combination of TODIM and TOPSIS method is used to adapt to more complex and changing circumstance. Considering the DMs' risk attitudes and simplifying the calculation process, a new method is proposed as follows:

Assume the possible solutions collection $AL = (AL_1, AL_2, \dots, AL_m)$ and attributes collection $AT = (AT_1, AT_2, \dots, AT_n)$, the decision matrix is denoted as $M = [r_{xy}(q_{xy})]_{m \times n}$. The weighting vector is denoted as $w_y = (w_1, w_2, \dots, w_n)$. The process of this new method in PHFSs is shown as follows:

Step 1: Standardize the decision matrix $M^* = [r_{xy}^* (q_{xy}^*)]_{m \times n}$ using Equations (20), (21)

Transform the cost attribute into benefit attribute by the following equations:

When AT_y is the positive attribute, the normalized PHFE is determined by Equation (20)

$$r_{xy}^* = \{m_{xy} (q_{xy})\} \quad (20)$$

When AT_y is the negative attribute, the normalized PHFE is determined by Equation (21)

$$r_{xy}^* = \{(1 - m_{xy}) (q_{xy})\} \quad (21)$$

Then normalize the decision matrices using the steps in Definition 3.

Step 2: Determine the positive and negative ideal solution $(r_y^+ (q_y^+))$ and $(r_y^- (q_y^-))$ indexes by the following equation:

$$\begin{cases} r_y^+ (q_y^+) = \max_{y \in S} (r_{xy}^*) \\ r_y^- (q_y^-) = \min_{y \in S} (r_{xy}^*) \end{cases} \quad (22)$$

Step 3: calculate the relative weight by Equation (23)

$$w_y^* = \frac{w_y}{\ddot{w}} \quad (23)$$

where $\ddot{w}$ denotes the maximum value of the attributes' weight.

Step 4: By referring to prospect theory, the PIS and NIS is regarded as the preference solution. The cost function is determined by Equation (24)

$$\begin{cases} \vartheta_{xy}^+ = \sqrt{\frac{w_y^*}{\sum_{y=1}^n w_y^*} \cdot d_y^- (r_{xy}^*, r_{xy}^+)} \\ \vartheta_{xy}^- = -\frac{1}{\theta} \sqrt{\frac{\sum_{y=1}^n w_y^*}{w_y^*} \cdot d_y^+ (r_{xy}^*, r_{xy}^+)} \end{cases} \quad (24)$$

Step 5: Finally, calculate the overall degree φ_x

$$\varphi_x = \frac{\sum_{y=1}^n (\vartheta_{xy}^+ + \vartheta_{xy}^-) - \min_{1 \leq t \leq m} \sum_{y=1}^n (\vartheta_{ty}^+ + \vartheta_{ty}^-)}{\max_{1 \leq t \leq m} \sum_{y=1}^n (\vartheta_{ty}^+ + \vartheta_{ty}^-) - \min_{1 \leq t \leq m} \sum_{y=1}^n (\vartheta_{ty}^+ + \vartheta_{ty}^-)} \quad (25)$$

Step 6: the final ranking results are ranked by the overall degree φ_x .

Obviously, the proposed method reduces the distance calculation between two possible solutions. Based on the TODIM and TOPSIS method, this method not only takes the bounded rationality of DMs, but also simplify the calculation process. Therefore, considering the simplicity of this method and the complement of TODIM and TOPSIS method.

4. An example and comparison

4.1. An example study for quality evaluation of college student employment and entrepreneurship education

The State Council attaches great importance to the employment and entrepreneurship work of college graduates. Since 2014, a notice has been issued every year to clarify the policy guidance for college graduates' employment and entrepreneurship, propose the implementation of the "College Student Entrepreneurship Guidance Plan" [2], and issue an annual report on the employment quality of college graduates to promote the dual increase of the proportion of college graduates' employment and entrepreneurship. In May 2019, Li Keqiang made important instructions on the national employment and entrepreneurship work and the employment and entrepreneurship work of ordinary college graduates, providing strong support for national college graduates to achieve higher quality and more full employment and entrepreneurship. Vocational colleges are the main battlefield for cultivating technical and skilled talents and carrying out employment and entrepreneurship education. Employment provides a stage for entrepreneurship, and entrepreneurship creates conditions for students' career development. At present, continuously reforming and improving the content and mode of employment and entrepreneurship education in vocational colleges, achieving the organic unity of employment and entrepreneurship education in talent cultivation paths, and constructing an employment and entrepreneurship education system are of great practical significance for effectively improving students' employment and entrepreneurship abilities, cultivating innovative thinking and entrepreneurship concepts, and further cultivating students' sense of social responsibility and professional quality. The employment education of students mainly includes the understanding of career, career development, the establishment of correct employment outlook, the ability to compare and choose employers, job search strategies, interview skills, the correct handling of the relationship between immediate interests and long-term development, as well as professional ability, social ability, psychological quality, professional ethics, etc. The entrepreneurship education of students mainly includes the pursuit of careers, the desire to reflect life values, the spirit of dedication, innovation, entrepreneurship, meticulous, pragmatic, and constantly improving

Table 1
Decision matrix M^1

	AT_1	AT_2
AL_1	{0.2 (0.3), 0.4 (0.7)}	{0.7 (1)}
AL_2	{0.3 (0.5), 0.6 (0.5)}	{0.8 (0.5), 0.5 (0.5)}
AL_3	{0.6 (0.1), 0.7 (0.5), 0.8 (0.4)}	{0.5 (0.1), 0.4 (0.4), 0.3 (0.5)}
AL_4	{0.2 (0.5), 0.4 (0.2), 0.3 (0.3)}	{0.7 (0.5), 0.8 (0.5)}
AL_5	{0.3 (1)}	{0.7 (1)}
	AT_3	AT_4
AL_1	{0.2 (1)}	{0.3 (0.2), 0.4 (0.8)}
AL_2	{0.4 (0.5), 0.3 (0.5)}	{0.2 (0.4), 0.5 (0.6)}
AL_3	{0.7 (1)}	{0.3 (0.3), 0.6 (0.7)}
AL_4	{0.3 (0.4), 0.2 (0.6)}	{0.3 (0.4), 0.4 (0.6)}
AL_5	{0.3 (0.4), 0.5 (0.3), 0.4 (0.3)}	{0.3 (0.5), 0.4 (0.5)}

Table 2
Decision matrix M^2

	AT_1	AT_2
AL_1	{0.3 (0.4), 0.4 (0.3), 0.5 (0.3)}	{0.7 (0.1), 0.6 (0.5), 0.3 (0.4)}
AL_2	{0.4 (0.1), 0.5 (0.6), 0.2 (0.3)}	{0.7 (0.5), 0.6 (0.5)}
AL_3	{0.7 (0.4), 0.8 (0.6)}	{0.6 (0.2), 0.3 (0.8)}
AL_4	{0.4 (0.4), 0.5 (0.6)}	{0.7 (0.2), 0.8 (0.4), 0.6 (0.4)}
AL_5	{0.1 (0.4), 0.3 (0.6)}	{0.7 (1)}
	AT_3	AT_4
AL_1	{0.3 (0.5), 0.5 (0.5)}	{0.2 (0.1), 0.3 (0.5), 0.4 (0.4)}
AL_2	{0.4 (1)}	{0.3 (0.2), 0.4 (0.6), 0.5 (0.2)}
AL_3	{0.5 (0.2), 0.6 (0.6), 0.7 (0.2)}	{0.4 (0.2), 0.7 (0.5), 0.8 (0.3)}
AL_4	{0.3 (0.2), 0.4 (0.5), 0.5 (0.3)}	{0.4 (0.1), 0.3 (0.5), 0.2 (0.4)}
AL_5	{0.2 (0.3), 0.3 (0.5), 0.4 (0.2)}	{0.4 (1)}

Table 3
Decision matrix M^3

	AT_1	AT_2
AL_1	{0.2 (0.4), 0.3 (0.6)}	{0.6 (1)}
AL_2	{0.2 (0.2), 0.5 (0.5), 0.1 (0.3)}	{0.7 (0.6), 0.6 (0.4)}
AL_3	{0.5 (0.4), 0.7 (0.6)}	{0.4 (0.1), 0.73 (0.6), 0.4 (0.3)}
AL_4	{0.3 (0.5), 0.2 (0.5)}	{0.8 (0.5), 0.7 (0.5)}
AL_5	{0.4 (0.2), 0.5 (0.4), 0.2 (0.4)}	{0.7 (0.5), 0.6 (0.5)}
	AT_3	AT_4
AL_1	{0.3 (0.4), 0.5 (0.2), 0.4 (0.4)}	{0.4 (0.5), 0.3 (0.5)}
AL_2	{0.3 (0.5), 0.2 (0.5)}	{0.3 (0.2), 0.5 (0.5), 0.4 (0.3)}
AL_3	{0.7 (0.3), 0.6 (0.7)}	{0.5 (0.1), 0.6 (0.5), 0.7 (0.4)}
AL_4	{0.2 (0.5), 0.4 (0.5)}	{0.3 (1)}
AL_5	{0.4 (0.3), 0.3 (0.7)}	{0.2 (0.3), 0.4 (0.3), 0.3 (0.4)}

qualities, social adaptability, innovation, and independent entrepreneurship abilities. The State Council attaches great importance to the employment and entrepreneurship work of college graduates. Since 2014, a notice has been issued every year to clarify the policy guidance for college graduates' employment and entrepreneurship, propose the implementation of the "College Student Entrepreneurship Guidance Plan", and issue an annual report on the employment quality of college graduates, in order to promote the dual increase in the proportion of college graduates' employment and entrepreneurship. In May

2019, Li Keqiang made important instructions on the national employment and entrepreneurship work and the employment and entrepreneurship work of ordinary college graduates, providing strong support for national college graduates to achieve higher quality and more full employment and entrepreneurship. Local comprehensive universities are the main battlefield for cultivating technical and skilled talents and carrying out employment and entrepreneurship education. Employment provides a stage for entrepreneurship, and entrepreneurship creates conditions for students' career development. At present,

Table 4
Decision matrix M^1

	AT_1	AT_2
AL_1	{0.2 (0.1), 0.2 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.4 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_2	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.6 (0.2), 0.6 (0.3)}	{0.2 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.5 (0.2), 0.5 (0.3)}
AL_3	{0.6 (0.1), 0.8 (0.2), 0.8 (0.2) # 0.7 (0.2), 0.7 (0.3)}	{0.5 (0.1), 0.6 (0.2), 0.6 (0.2) # 0.7 (0.2), 0.7 (0.3)}
AL_4	{0.2 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.4 (0.2), 0.3 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.2 (0.2), 0.2 (0.3)}
AL_5	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.3 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.3 (0.3)}
	AT_3	AT_4
AL_1	{0.2 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.2 (0.2), 0.2 (0.3)}	{0.4 (0.1), 0.3 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_2	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.3 (0.3)}	{0.5 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.5 (0.2), 0.5 (0.3)}
AL_3	{0.7 (0.1), 0.7 (0.2), 0.7 (0.2) # 0.7 (0.2), 0.7 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.6 (0.2) # 0.6 (0.2), 0.6 (0.3)}
AL_4	{0.3 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.2 (0.2), 0.3 (0.3)}	{0.4 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_5	{0.5 (0.1), 0.5 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.4 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}

Table 5
Decision matrix M^2

	AT_1	AT_2
AL_1	{0.4 (0.1), 0.4 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.5 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_2	{0.4 (0.1), 0.5 (0.2), 0.5 (0.2) # 0.5 (0.2), 0.2 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_3	{0.7 (0.1), 0.8 (0.2), 0.8 (0.2) # 0.8 (0.2), 0.7 (0.3)}	{0.7 (0.1), 0.4 (0.2), 0.7 (0.2) # 0.7 (0.2), 0.7 (0.3)}
AL_4	{0.4 (0.1), 0.5 (0.2), 0.5 (0.2) # 0.5 (0.2), 0.4 (0.3)}	{0.2 (0.1), 0.3 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.2 (0.3)}
AL_5	{0.3 (0.1), 0.1 (0.2), 0.1 (0.2) # 0.3 (0.2), 0.3 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.3 (0.3)}
	AT_3	AT_4
AL_1	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.5 (0.2), 0.5 (0.3)}	{0.2 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_2	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.4 (0.3)}	{0.4 (0.1), 0.3 (0.2), 0.5 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_3	{0.6 (0.1), 0.5 (0.2), 0.7 (0.2) # 0.6 (0.2), 0.6 (0.3)}	{0.8 (0.1), 0.8 (0.2), 0.4 (0.2) # 0.7 (0.2), 0.7 (0.3)}
AL_4	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.5 (0.3)}	{0.4 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_5	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.2 (0.3)}	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.4 (0.3)}

continuous reform and improvement of the content and mode of employment and entrepreneurship education in local comprehensive universities, the organic unity of employment and entrepreneurship education in talent cultivation paths, and the construction of an employment and entrepreneurship education system are of great practical significance for effectively improving students' employment and entrepreneurship abilities, cultivating innova-

tive thinking and entrepreneurship concepts, and further cultivating students' sense of social responsibility and professional quality. The employment education of students mainly includes the understanding of career, career Career development, the establishment of correct employment outlook, the ability to compare and choose employers, job search strategies, interview skills, the correct handling of the relationship between immediate interests

Table 6
Decision matrix M^3

	AT_1	AT_2
AL_1	{0.2 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.2 (0.3)}	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.4 (0.3)}
AL_2	{0.5 (0.1), 0.5 (0.2), 0.5 (0.2) # 0.2 (0.2), 0.1 (0.3)}	{0.3 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_3	{0.5 (0.1), 0.7 (0.2), 0.7 (0.2) # 0.7 (0.2), 0.5 (0.3)}	{0.6 (0.1), 0.7 (0.2), 0.7 (0.2) # 0.7 (0.2), 0.6 (0.3)}
AL_4	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.2 (0.2), 0.2 (0.3)}	{0.2 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_5	{0.5 (0.1), 0.4 (0.2), 0.5 (0.2) # 0.5 (0.2), 0.5 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}
	AT_3	AT_4
AL_1	{0.3 (0.1), 0.5 (0.2), 0.4 (0.2) # 0.4 (0.2), 0.3 (0.3)}	{0.4 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_2	{0.5 (0.1), 0.5 (0.2), 0.5 (0.2) # 0.3 (0.2), 0.4 (0.3)}	{0.3 (0.1), 0.4 (0.2), 0.4 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_3	{0.5 (0.1), 0.7 (0.2), 0.7 (0.2) # 0.6 (0.2), 0.6 (0.3)}	{0.6 (0.1), 0.7 (0.2), 0.7 (0.2) # 0.7 (0.2), 0.6 (0.3)}
AL_4	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.3 (0.3)}	{0.2 (0.1), 0.2 (0.2), 0.2 (0.2) # 0.3 (0.2), 0.3 (0.3)}
AL_5	{0.2 (0.1), 0.2 (0.2), 0.3 (0.2) # 0.3 (0.2), 0.4 (0.3)}	{0.3 (0.1), 0.3 (0.2), 0.3 (0.2) # 0.4 (0.2), 0.4 (0.3)}

Table 7
Decision matrix M^*

	AT_1	AT_2
AL_1	{0.29 (0.1), 0.31 (0.2), 0.33 (0.2) # 0.33 (0.2), 0.39 (0.3)}	{0.33 (0.1), 0.33 (0.2), 0.33 (0.2) # 0.37 (0.2), 0.37 (0.3)}
AL_2	{0.41 (0.1), 0.45 (0.2), 0.45 (0.2) # 0.46 (0.2), 0.33 (0.3)}	{0.27 (0.1), 0.30 (0.2), 0.30 (0.2) # 0.41 (0.2), 0.41 (0.3)}
AL_3	{0.62 (0.1), 0.77 (0.2), 0.77 (0.2) # 0.74 (0.2), 0.65 (0.3)}	{0.62 (0.1), 0.57 (0.2), 0.67 (0.2) # 0.70 (0.2), 0.67 (0.3)}
AL_4	{0.31 (0.1), 0.36 (0.2), 0.36 (0.2) # 0.39 (0.2), 0.31 (0.3)}	{0.23 (0.1), 0.27 (0.2), 0.31 (0.2) # 0.31 (0.2), 0.23 (0.3)}
AL_5	{0.37 (0.1), 0.26 (0.2), 0.30 (0.2) # 0.37 (0.2), 0.37 (0.3)}	{0.30 (0.1), 0.30 (0.2), 0.30 (0.2) # 0.33 (0.2), 0.33 (0.3)}
	AT_3	AT_4
AL_1	{0.27 (0.1), 0.34 (0.2), 0.30 (0.2) # 0.39 (0.2), 0.36 (0.3)}	{0.33 (0.1), 0.37 (0.2), 0.40 (0.2) # 0.33 (0.2), 0.33 (0.3)}
AL_2	{0.43 (0.1), 0.43 (0.2), 0.43 (0.2) # 0.34 (0.2), 0.37 (0.3)}	{0.41 (0.1), 0.30 (0.2), 0.39 (0.2) # 0.41 (0.2), 0.41 (0.3)}
AL_3	{0.61 (0.1), 0.63 (0.2), 0.70 (0.2) # 0.63 (0.2), 0.63 (0.3)}	{0.64 (0.1), 0.67 (0.2), 0.57 (0.2) # 0.67 (0.2), 0.64 (0.3)}
AL_4	{0.34 (0.1), 0.31 (0.2), 0.31 (0.2) # 0.27 (0.2), 0.39 (0.3)}	{0.35 (0.1), 0.23 (0.2), 0.23 (0.2) # 0.33 (0.2), 0.33 (0.3)}
AL_5	{0.34 (0.1), 0.34 (0.2), 0.30 (0.2) # 0.4 (0.2), 0.33 (0.3)}	{0.34 (0.1), 0.34 (0.2), 0.34 (0.2) # 0.40 (0.2), 0.40 (0.3)}

and long-term development, as well as professional ability, social ability, psychological quality, professional ethics, etc. The entrepreneurship education of students mainly includes the pursuit of careers, the desire to reflect life values, the spirit of dedication, innovation, entrepreneurship, meticulous, pragmatic, and constantly improving qualities, social adaptability, innovation, and independent

entrepreneurship abilities. The ability and level of employment and entrepreneurship guidance teachers in local comprehensive universities directly affect students' employment and entrepreneurship abilities. We should organize full-time and part-time teachers for employment and entrepreneurship guidance to cultivate their ability to guide employment and entrepreneurship through learning, training, practice,

and competitions; Arrange teachers to go out in a planned manner, broaden their horizons, communicate with the outside world, observe and learn from the specific practices, advanced experiences, and concepts of employment and entrepreneurship education for college students in other universities, inspire teachers to guide their thinking on employment and entrepreneurship, and thus improve their own level. Local comprehensive universities should further deepen the reform of their personnel system, select individuals with experience in employment guidance, innovation and entrepreneurship, winning awards in entrepreneurship and entrepreneurship competitions, and starting enterprises to serve as guidance teachers for employment and entrepreneurship education in universities. Senior experts and human resource managers of enterprises should be invited to the university to hold employment and entrepreneurship lectures, and universities should form a team of science and technology innovation teachers with teaching departments, and carry out employment and entrepreneurship training based on students' majors. Local comprehensive universities should cultivate students to think and design entrepreneurial projects based on their own professional knowledge, and guide work such as physical production, thematic research, and topic exploration to strengthen students' entrepreneurial ability training. Drive students' practice through activities organized by technology clubs, such as organizing innovation competitions based on professional knowledge, cultivating students' ability to capture inspiration, analyze, solve problems, and seek breakthroughs; Utilizing innovative achievements to drive employment and entrepreneurship, such as establishing entrepreneurship nurseries on campus, selecting excellent projects to enter the school, providing incubation venues for student entrepreneurship projects, providing certain policy support, promoting project implementation and transformation, and driving student employment and entrepreneurship [3]; Utilize collaborative innovation and entrepreneurship centers to provide corresponding scientific research rewards to team members or teachers who have successfully entered the school. Local comprehensive universities should further carry out employment and entrepreneurship competitions and cultivate a school level guidance team of employment and entrepreneurship teachers; Each department carries out innovative and creative works competition by specialty, and establishes Student society to guide and cultivate students' employment and entrepreneurship

Table 8
The positive distance $d_y^+ (r_{xy}^*, r_{xy}^+)$

	AT_1	AT_2	AT_3	AT_4
AL_1	0.36	0.30	0.31	0.29
AL_2	0.26	0.31	0.23	0.25
AL_3	0.00	0.00	0.00	0.00
AL_4	0.34	0.41	0.33	0.37
AL_5	0.36	0.35	0.32	0.27

Table 9
The negative distance $d_y^- (r_{xy}^*, r_{xy}^+)$

	AT_1	AT_2	AT_3	AT_4
AL_1	0.06	0.12	0.04	0.10
AL_2	0.13	0.11	0.10	0.13
AL_3	0.36	0.41	0.32	0.37
AL_4	0.08	0.00	0.04	0.00
AL_5	0.00	0.08	0.00	0.12

Table 10
The positive cost function ϑ_{xy}^+

	AT_1	AT_2	AT_3	AT_4
AL_1	0.12	0.17	0.12	0.13
AL_2	0.18	0.16	0.18	0.15
AL_3	0.31	0.30	0.34	0.25
AL_4	0.14	0.00	0.13	0.00
AL_5	0.00	0.13	0.00	0.14

ability. Promote and test the effectiveness of employment and entrepreneurship education in universities by participating in competitions and projects. The quality evaluation of college student employment and entrepreneurship education is regarded as the defined MAGDM. Based on the above analysis, this process can be regarded as MAGDM problems. There are five possible local comprehensive universities $AL = (AL_1, AL_2, \dots, AL_5)$ to select. The decision experts select four defined attributes $AT = (AT_1, AT_2, AT_3, AT_4)$ to assess five local comprehensive universities: ① AT_1 is the employment and entrepreneurship management system; ② AT_2 is the employment and entrepreneurship costs; ③ AT_3 is the allocation of employment and entrepreneurship resources; ④ AT_4 is the integrated education and industry education system. Evidently, AT_2 is the cost. The constructed decision matrix is denoted as in Tables 1–3. Besides, the weighting vector is denoted as $w_c = (w_1, w_2, \dots, w_l)$ which can be calculated by entropy method. The weighting ration among DMs is denoted as $e = (e_1, e_2, e_3)^T = (0.3, 0.4, 0.3^T)$. Then, TODIM-TOPSIS under PHFSs is built to solve MAGDM for quality evaluation of college student employment and entrepreneurship education.

Table 11
The negative cost function ϑ_{xy}^-

	AT_1	AT_2	AT_3	AT_4
AL_1	-0.53	-0.51	-0.42	-0.59
AL_2	-0.45	-0.52	-0.36	-0.56
AL_3	0.00	0.00	0.00	0.00
AL_4	-0.51	-0.60	-0.43	-0.68
AL_5	-0.53	-0.55	-0.42	-0.58

Step 1: Obtain the standardized the matrix, and the normalized matrices $M_c^* = [r_{c\ xy}^* (r_c^*)]_{m \times n}$ are shown in Tables 4–6.

Step 2: Aggregate decision matrices into one overall decision matrix (See Table 7).

Step 3: The given weight information is as follows:

$$w_y = \{0.258, 0.227, 0.354, 0.161\}$$

Step 4: Figure out the relative weight w_y^*

$$w_y^* = \{0.728, 0.642, 1, 0.456\}$$

Step 5: Determine the positive and negative ideal solution $(r^+{}_y (q^+{}_y) \text{ and } r^-{}_y (q^-{}_y))$ indexes

$$\begin{aligned} r^+{}_y (q^+{}_y) &= \{0.62 (0.1), 0.77 (0.2), 0.77 (0.2), 0.74 (0.2), 0.65 (0.3)\} \# \\ &\{0.62 (0.1), 0.57 (0.2), 0.67 (0.2), 0.70 (0.2), 0.67 (0.3)\} \\ &\{0.61 (0.1), 0.63 (0.2), 0.70 (0.2), 0.63 (0.2), 0.63 (0.3)\} \\ &\{0.62 (0.1), 0.57 (0.2), 0.67 (0.2), 0.70 (0.2), 0.67 (0.3)\} \end{aligned}$$

$$\begin{aligned} r^-{}_y (q^-{}_y) &= \{0.37 (0.1), 0.26 (0.2), 0.30 (0.2), 0.37 (0.2), 0.37 (0.3)\} \# \\ &\{0.23 (0.1), 0.27 (0.2), 0.31 (0.2), 0.31 (0.2), 0.23 (0.3)\} \\ &\{0.34 (0.1), 0.34 (0.2), 0.30 (0.2), 0.34 (0.2), 0.33 (0.3)\} \\ &\{0.34 (0.1), 0.34 (0.2), 0.34 (0.2), 0.40 (0.2), 0.47 (0.3)\} \end{aligned}$$

Step 6: Calculate the positive and negative distance (See Tables 8, 9).

Step 7: By referring to prospect theory, the PIS and NIS is regarded as the preference solution. The cost function $\vartheta_{xy}^{+(-)}$ is shown in Tables 10, 11.

$$\begin{cases} \vartheta_{xy}^+ = \sqrt{\frac{w_y^*}{\sum_{y=1}^n w_y^*}} \cdot d_y^- (r_{xy}^*, r_{xy}^+) \\ \vartheta_{xy}^- = -\frac{1}{\theta} \sqrt{\frac{\sum_{y=1}^n w_y^*}{w_y^*}} \cdot d_y^+ (r_{xy}^*, r_{xy}^+) \end{cases}$$

Step 8: Finally, calculate the overall degree φ_x

$$\varphi_x = \{0.140, 0.232, 1, 0, 0.044\}$$

Step 9: The final ranking results are ranked by the overall degree φ_x .

$$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$$

4.2. Comparative analysis

In this part, the TODIM-TOPSIS method under PHFSs is compared with some existing methods, such as, PHFWA operators [65], PHFWG operators [65], PHF-TODIM method [65], PHF-CODAS method [68], PHF-EDAS method [69], PHF-TOPSIS method [70], OHF-COPRAS method [71] and PHF-GLDS method [72]. Then, the results of these different methods are depicted in Table 12.

Derived from the Table 12, it is evident that the best local comprehensive university is AL_3 , while the worst local comprehensive university is AL_4 . In other words, nine methods' order is slightly different. All these decision methods can effectively cope with MAGDM from different decision angles. Furthermore, the TODIM-TOPSIS method under PHFSs combined the main advantages of TODIM and TOPSIS. Compared with the existing methods, the TODIM-TOPSIS method under PHFSs has advantages in terms of practical decision-making dependability and stability.

5. Conclusion

Deepening the reform of innovation and entrepreneurship education in higher education institutions is an urgent need for the country to implement the innovation driven development strategy and promote the upgrading of economic quality and efficiency. It is an important measure to promote the comprehensive reform of higher education and promote higher quality entrepreneurship and employment for college graduates. The innovation and entrepreneurship competition, as an effective carrier for deepening the reform of innovation and entrepreneurship education, is an important platform for promoting the comprehensive development of college students, and a bridge and link to improve the competitiveness of college students in employment and lead them to achieve higher quality and full employment. In recent years, Ludong University has encouraged students to participate in innovation and entrepreneurship competitions. As a starting point for implementing innovation and entrepreneurship education, it has created a new path of competition

Table 12
Order results of different methods

Methods	Ranking order	The optimal alternative
PHFWA operators [65]	$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$	AL_3
PHFWG operators [65]	$AL_3 > AL_2 > AL_5 > AL_1 > AL_4$	AL_3
PHF-TODIM method [65]	$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$	AL_3
PHF-CODAS method [68]	$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$	AL_3
PHF-EDAS method [69]	$AL_3 > AL_2 > AL_5 > AL_1 > AL_4$	AL_3
PHF-TOPSIS method [70]	$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$	AL_3
PHF-COPRAS method [71]	$AL_3 > AL_2 > AL_5 > AL_1 > AL_4$	AL_3
PHF-GLDS method [72]	$AL_3 > AL_2 > AL_1 > AL_5 > AL_4$	AL_3

education, improved students' competitiveness in entrepreneurship and employment, and led students to achieve higher quality and full employment. The quality evaluation of college student employment and entrepreneurship education is regarded as the defined MAGDM. As a set to extend the fuzzy sets, the PHFSs provides a more flexible and effective method to describe the decision-making environment. The classical TODIM method deriving from the PT, which considers the psychological factor of DMs, and the principle of TOPSIS method is to evaluate the distance between ideal solution. In this paper, the TODIM-TOPSIS method in PHFSs not only combines the unique points of two traditional methods, but also simplifies the original method. at last, an example for quality evaluation of college student employment and entrepreneurship education is to set to prove this method's practicability and feasibility. In the future plans, the cumulative prospect theory is supposed to added to improve this new method, and is also supposed to be more scientific fuzzy sets.

References

- [1] H. Garg, Z. Ali, I.M. Hezam and J. Gwak, Decision-making approach based on generalized aggregation operators with complex single-valued neutrosophic hesitant fuzzy set information, *Mathematical Problems in Engineering* **2022** (2022), 20.
- [2] H. Garg and G. Kaur, Algorithm for solving the decision-making problems based on correlation coefficients under cubic intuitionistic fuzzy information: a case study in watershed hydrological system, *Complex & Intelligent Systems* **8** (2022), 179–198.
- [3] H. Garg and D. Rani, An efficient intuitionistic fuzzy MULTIMOORA approach based on novel aggregation operators for the assessment of solid waste management techniques, *Applied Intelligence* **52** (2022), 4330–4363.
- [4] H. Garg and D. Rani, Novel distance measures for intuitionistic fuzzy sets based on various triangle centers of isosceles triangular fuzzy numbers and their applications, *Expert Systems with Applications* **191** (2022), 20.
- [5] H. Hashim, H. Garg, A. Al-Quran, N.A. Awang and L. Abdullah, Heronian mean operators considering shapley fuzzy measure under interval neutrosophic vague environment for an investment decision, *International Journal of Fuzzy Systems* **24** (2022), 2068–2091.
- [6] F.F. Jin, Y. Zhang, H. Garg, J.P. Liu and J. Chen, Evaluation of small and medium-sized enterprises' sustainable development with hesitant fuzzy linguistic group decision-making method, *Applied Intelligence* **52** (2022), 4940–4960.
- [7] B. Chen, Q. Cai, G. Wei and Z. Mo, Novel Aczel-Alsina operations-based linguistic Z-number aggregation operators and their applications in multi-attribute group decision-making process, *Engineering Applications of Artificial Intelligence* **124** (2023), 106541.
- [8] H. Sun, Z. Yang, Q. Cai, G.W. Wei and Z.W. Mo, An extended Exp-TODIM method for multiple attribute decision making based on the Z-Wasserstein distance, *Expert Systems with Applications* **214** (2023), 14.
- [9] H.Y. Zhang and G.W. Wei, Location selection of electric vehicles charging stations by using the spherical fuzzy CPT-CoCoSo and D-CRITIC method, *Computational & Applied Mathematics* **42** (2023), 35.
- [10] J. Campbell, H. Ahmad, S. Claffin, B. Taylor, I. van der Mei, R. Norman, G. Chen, L. Laslett, S. Simpson-Yap, G. Henson and A. Palmer, A multi-instrument comparison study of six multi-attribute utility instruments for a large cohort of Australians with multiple sclerosis: implications for resource allocation decision making, *Qual. Life Res.* **30** (2021), S45–S46.
- [11] T. Mahmood, W. Ali, Z. Ali and R. Chinram, Power aggregation operators and similarity measures based on improved intuitionistic hesitant fuzzy sets and their applications to multiple attribute decision making, *Cmes-Computer Modeling in Engineering & Sciences* **126** (2021), 1165–1187.
- [12] M.B. Maisuria, D.M. Sonar and M.K. Rathod, Nanofluid selection used for coolant in heat exchanger by multiple attribute decision-making method, *Journal of Mechanical Science and Technology* **35** (2021), 689–695.
- [13] A. Mousazadeh, M. Kafaee and M. Ashraf, Ranking of commercial photodiodes in radiation detection using multiple-attribute decision making approach, *Nucl. Instrum. Methods Phys. Res. Sect. A-Accel. Spectrom. Dect. Assoc. Equip.* **987** (2021), 5.
- [14] G. Shahzadi, F. Zafar and M.A. Alghamdi, Multiple-attribute decision-making using fermatean fuzzy hamacher interactive geometric operators, *Mathematical Problems in Engineering* **2021** (2021), 20.
- [15] A. Varmaghani, A.M. Nazar, M. Ahmadi, A. Sharifi, S.J. Ghouschi and Y. Pourasad, DMTC: optimize energy consumption in dynamic wireless sensor network based on fog computing and fuzzy multiple attribute decision-making, *Wireless Communications & Mobile Computing*

- 2021** (2021), 14.
- [16] H. Liang, C.-C. Li, Y. Dong and F. Herrera, Linguistic opinions dynamics based on personalized individual semantics, *IEEE Transactions on Fuzzy Systems* **29** (2021), 2453–2466.
 - [17] Y. Wu, Y. Dong, J. Qin and W. Pedrycz, Linguistic distribution and priority-based approximation to linguistic preference relations with flexible linguistic expressions in decision making, *IEEE Transactions on Cybernetics* **51** (2021), 649–659.
 - [18] S. Zhao, Y. Dong, S. Wu and L. Martinez, Linguistic scale consistency issues in multi-granularity decision making contexts, *Applied Soft Computing* **101** (2021).
 - [19] H. Zhang, W. Zhu, X. Chen, Y. Wu, H. Liang, C.-C. Li and Y. Dong, Managing flexible linguistic expression and ordinal classification-based consensus in large-scale multi-attribute group decision making, *Annals of Operations Research* (2022).
 - [20] H. Zhang, X. Wang, W. Xu and Y. Dong, From numerical to heterogeneous linguistic best-worst method: Impacts of personalized individual semantics on consistency and consensus, *Engineering Applications of Artificial Intelligence* **117** (2023).
 - [21] L.A. Zadeh, Fuzzy sets, *Information and Control* **8** (1965), 338–356.
 - [22] T.Y. Chen, An interval-valued pythagorean fuzzy outranking method with a closeness-based assignment model for multiple criteria decision making, *International Journal of Intelligent Systems* **33** (2018), 126–168.
 - [23] H. Garg, New exponential operational laws and their aggregation operators for interval-valued Pythagorean fuzzy multicriteria decision-making, *International Journal of Intelligent Systems* **33** (2018), 653–683.
 - [24] S.D. Xian, Y. Xiao, Z.J. Yang, Y.H. Li and Z.Y. Han, A new trapezoidal Pythagorean fuzzy linguistic entropic combined ordered weighted averaging operator and its application for enterprise location, *International Journal of Intelligent Systems* **33** (2018), 1880–1899.
 - [25] S.Z. Zeng, C.D. Cao, Y. Deng and X.D. Shen, Pythagorean fuzzy information aggregation based on weighted induced operator and its application to R&D projections selection, *Informatica* **29** (2018), 567–580.
 - [26] K.T. Atanassov, Intuitionistic fuzzy sets, *Fuzzy Sets and Systems* **20** (1986), 87–96.
 - [27] V. Torra and Y. Narukawa, IEEE, *On Hesitant Fuzzy Sets and Decision* (2009).
 - [28] Z.S. Xu and W. Zhou, Consensus building with a group of decision makers under the hesitant probabilistic fuzzy environment, *Fuzzy Optimization and Decision Making* **16** (2017), 481–503.
 - [29] J. Guo, J.L. Yin, L. Zhang, Z.F. Lin and X. Li, Extended TODIM method for CCUS storage site selection under probabilistic hesitant fuzzy environment, *Applied Soft Computing* **93** (2020), 106381.
 - [30] Z.Y. Chen, D. Shen, Y.J. Ren, F.P. Yu and X.J. Yuan, Airspace operation effectiveness evaluation based on q-rung orthopair probabilistic hesitant fuzzy GRA and TOPSIS, *Symmetry-Basel* **14** (2022), 242.
 - [31] Z.Q. Li, VIKOR method for evaluating the management quality of teacher education with probabilistic hesitant fuzzy information, *Discrete Dynamics in Nature and Society* **2021** (2021), 1326202.
 - [32] G.C. Abiogu, Entrepreneurship education and employment creation in nigeria: a philosophical analysis, *J. Home Econ. Res.* **9** (2008), 19–26.
 - [33] W.-L. Jin and B.-F. Lin, A practical example on enhancing employability and entrepreneurship in institutions of higher education in China, in: 2nd International Conference on Society and Information Technologies, ICSIT 2011, March 27 2011 – March 30, 2011, International Institute of Informatics and Systemics, IIIS, Orlando, FL, United states (2011), 182–187.
 - [34] S.J.F. Hosseini, R. Nadafi and F. Lashgarara, Entrepreneurship education and employment in agricultural sector of Iran, *Tech. Technol. Educ. Manag.-TTEM* **7** (2012), 1413–1417.
 - [35] J. Huang, Analyzing elements of the employment and entrepreneurship practice education of college students, *BioTechnology: An Indian Journal* **10** (2014), 3521–3526.
 - [36] X. Yao and L.C. He, Problems existing in entrepreneurship education in colleges and universities under the new situation of employment and discussion on countermeasures, in: International Conference on Management, Information and Educational Engineering, MIEE 2014, November 22, 2014 – November 23, 2014, CRC Press/Balkema, Xiamen, China (2015), 605–610.
 - [37] B. Akuegwu, Perceived effectiveness of south-south nigerian public universities in managing entrepreneurship education for employment creation, in: 10th International Technology, Education and Development Conference (INTED), Iated-Int Assoc Technology Education & Development, Valencia, SPAIN (2016), 8561–8570.
 - [38] P. Premand, S. Brodmann, R. Almeida, R. Grun and M. Barouni, Entrepreneurship Education and Entry into Self-Employment Among University Graduates, *World Dev.* **77** (2016), 311–327.
 - [39] X.Y. Zhong, Social-adaptation-ability-enhancement-oriented Employment and Entrepreneurship Education of University Students, in: 3rd International Conference on Economic, Business Management and Education Innovation (EBMEI 2016), Singapore Management & Sports Science Inst Pte Ltd, Prague, CZECH REPUBLIC (2016), 624–627.
 - [40] P.F. Kabore, Entrepreneurship, Education and Youth Employment in Africa: Reframing Higher Education, in: 1st International Conference on Innovation and Interdisciplinary Solutions for Underserved Areas (InterSol) / 6th National Colloquium on Computer Research and Applications (CNRIA), Springer, Dakar, SENEGAL (2017), 141–149.
 - [41] T. Lan and X.F. Zhang, Analysis on the Impact of Innovation and Entrepreneurship Education on the Employment Ability of Fashion Design Professionals, in: 7th International Conference on Education, Management, Information and Computer Science (ICEMC), Atlantis Press, Shenyang, PEOPLES R CHINA (2017), 1096–1100.
 - [42] J. Li, L. Ma, X. Wu and Z. Chen, Exploration on education practice based on employment and entrepreneurship in higher institutes of China, in: Springer Verlag (2017), 554–564.
 - [43] J. Zhao, J. Feng, M.H. Wu and I. Destech Publicat, Research on Curriculum Content and Education Model of Soft Ability Training of Employment and Entrepreneurship for Graduates of Science and Engineering, in: 2nd International Conference on Education, Management and Systems Engineering (EMSE), Destech Publications, Inc, Beijing, PEOPLES R CHINA (2017), 98–103.
 - [44] P.F. Kabore, Entrepreneurship, education and youth employment in Africa: Reframing higher education, in:

- 1st International Conference on Innovation and Interdisciplinary Solutions for Underserved Areas, InterSol 2017, and co-located with the 6th Colloquium National sur la Recherche en Informatique et ses Applications, CNRIA 2017, April 11, 2017 – April 12, 2017, Springer Verlag, Dakar, Senegal (2018), 141–149.
- [45] H. Hasleberg, K.H. Voldsund and S.T. Hagen, Entrepreneurship Education for Engineering Students – A Survey of Former Students' Self-Employment and Market Attraction, in: 10th IEEE Global Engineering Education Conference (EDUCON), IEEE, Dubai, U ARAB EMIRATES (2019), 337–344.
- [46] H. Hasleberg, K.H. Voldsund and S.T. Hagen, Entrepreneurship education for engineering students – A survey of former students' self-employment and market attraction, in: 10th IEEE Global Engineering Education Conference, EDUCON 2019, April 9, 2019 – April 11, 2019, IEEE Computer Society, Dubai, United Arab Emirates (2019), 337–344.
- [47] I. Otache, Entrepreneurship education and undergraduate students' self- and paid-employment intentions: A conceptual framework, *Educ. Train.* **61** (2019), 46–64.
- [48] C.-D. Anca, C.M. Alexandra and S. Adrian, Teaching Z Generation Engineers. Using Entrepreneurship Education to Develop Soft Skills and Match Employers' Expectations, in: 11th International Conference and Exposition on Electrical And Power Engineering, EPE 2020, October 22, 2020 – October 23, 2020, Institute of Electrical and Electronics Engineers Inc., Iasi, Romania (2020), 180–184.
- [49] W.W. Jiang, Exploration of College Students' Entrepreneurship and Employment Education under the Background of "Internet plus", *Basic Clin. Pharmacol. Toxicol.* **127** (2020), 173–173.
- [50] I. Otache, D.O. Oluwade and E.O.J. Idoko, Entrepreneurship education and undergraduate students' self-employment intentions: do paid employment intentions matter?, *Educ. Train.* **62** (2020), 741–757.
- [51] P.N. Chikasha, K. Ramdass, K. Mokgohloa and K.R. Ramdass, Balancing employment and entrepreneurship requirements in industrial engineering education, *S. Afr. J. Ind. Eng.* **32** (2021), 1–12.
- [52] J.L. Hu, H.M. Deng and M.L. Lai, Effect of mass entrepreneurship and innovation education on alleviating college students' employment anxiety, *Psychiatr. Danub.* **33** (2021), S37–S39.
- [53] C. Jia, J. Zuo and W. Lu, Influence of Entrepreneurship Education on Employment Quality and Employment Willingness, *Int. J. Emerg. Technol. Learn.* **16** (2021), 65–78.
- [54] W. Li, The influence of the integration of innovation and entrepreneurship education and professional education on the employment ability of vocational college students, in: 2nd International Conference on Computers, Information Processing and Advanced Education, CIPAE 2021, May 25, 2021 – May 27, 2021, Association for Computing Machinery, Ottawa, ON, Canada (2021), 314–318.
- [55] Z.Z. Sun and X.H. Dai, The influence of mass entrepreneurship and innovation education integrated into automobile specialty teaching on alleviating college students' employment psychological anxiety, *Psychiatr. Danub.* **33** (2021), S386–S389.
- [56] X. Wang, C. Hu, F. Fang, M. Han and X. Wang, Research on Employment and Entrepreneurship Education in Universities, in: 2nd Asia-Pacific Conference on Image Processing, Electronics and Computers, IPEC 2021, April 14, 2021 – April 16, 2021, Association for Computing Machinery, Dalian, China (2021), 712–715.
- [57] X. Wu, Discussion on the Organic Integration of Innovation and Entrepreneurship Education and Employment Guidance Education Under Internet+, in: International Conference on Multi-modal Information Analytics, MMIA 2021, April 23, 2021 – April 24, 2021, Springer Science and Business Media Deutschland GmbH, Virtual, Online (2021), 17–22.
- [58] S. Yin, W. Liu and Q. Yin, Research on the path of application-oriented undergraduate employment and entrepreneurship education under the background of industry-education integration, in: 2nd International Conference on Computers, Information Processing and Advanced Education, CIPAE 2021, May 25, 2021 – May 27, 2021, Association for Computing Machinery, Ottawa, ON, Canada (2021), 720–723.
- [59] I.K. Biney, Continuing education and employment creation: Investment in entrepreneurship matters, *J. Plan. Lit.* **37** (2022), 632–632.
- [60] F. Yang, Resource collection algorithm for entrepreneurship and employment education in universities based on data mining, *Mob. Inf. Syst.* **2022** (2022).
- [61] L.L. Zhao, The influence of ideological and political education reform on college students' employment and entrepreneurship anxiety, *Psychiatr. Danub.* **34** (2022), S777–S778.
- [62] V. Torra, Hesitant fuzzy sets, *International Journal of Intelligent Systems* **25** (2010), 529–539.
- [63] J. Li, Q.X. Chen, L.L. Niu and Z.X. Wang, An ORESTE approach for multi-criteria decision-making with probabilistic hesitant fuzzy information, *International Journal of Machine Learning and Cybernetics* **11** (2020), 1591–1609.
- [64] X. Sha, C. Yin, Z. Xu and S. Zhang, Probabilistic hesitant fuzzy TOPSIS emergency decision-making method based on the cumulative prospect theory, *Journal of Intelligent & Fuzzy Systems* **40** (2021), 4367–4383.
- [65] N.N. Liao, G.W. Wei and X.D. Chen, TODIM method based on cumulative prospect theory for multiple attributes group decision making under probabilistic hesitant fuzzy setting, *International Journal of Fuzzy Systems* **24** (2022), 322–339.
- [66] W.K. Zhang, J. Du and X.L. Tian, Finding a promising venture capital project with TODIM under Probabilistic Hesitant Fuzzy circumstance, *Technological and Economic Development of Economy* **24** (2018), 2026–2044.
- [67] J. Wu, X.D. Liu, Z.W. Wang and S.T. Zhang, Dynamic emergency decision-making method with probabilistic hesitant fuzzy information based on GM(1,1) and TOPSIS, *IEEE Access* **7** (2019), 7054–7066.
- [68] N.N. Liao, G.W. Wei, X.R. Xu, X.D. Chen and Y.F. Guo, CODAS method with probabilistic hesitant fuzzy information and its application to environmentally and economically balanced supplier selection, *Technological and Economic Development of Economy* **28** (2022), 1419–1438.
- [69] N.N. Liao, H. Gao, R. Lin, G.W. Wei and X.D. Chen, An extended EDAS approach based on cumulative prospect theory for multiple attributes group decision making with probabilistic hesitant fuzzy information, *Artificial Intelligence Review* **56** (2023), 2971–3003.
- [70] M. Naeem, M.A. Khan, S. Abdullah, M. Qiyas and S. Khan, Extended TOPSIS method based on the entropy measure and probabilistic hesitant fuzzy information and their application in decision support system, *Journal of Intelligent & Fuzzy Systems* **40** (2021), 11479–11490.
- [71] R. Krishankumar, H. Garg, K. Arun, A. Saha, K.S. Ravichandran and S. Kar, An integrated decision-making

- COPRAS approach to probabilistic hesitant fuzzy set information, *Complex & Intelligent Systems* **7** (2021), 2281–2298.
- [72] N.N. Liao, Q. Cai, H. Garg, G.W. Wei and X.R. Xu, Novel gained and lost dominance score method based on cumulative prospect theory for group decision-making problems in probabilistic hesitant fuzzy environment, *International Journal of Fuzzy Systems* (2023), <https://doi.org/10.1007/s40815-40022-01440-40817>.

Copyright of Journal of Intelligent & Fuzzy Systems is the property of IOS Press and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.